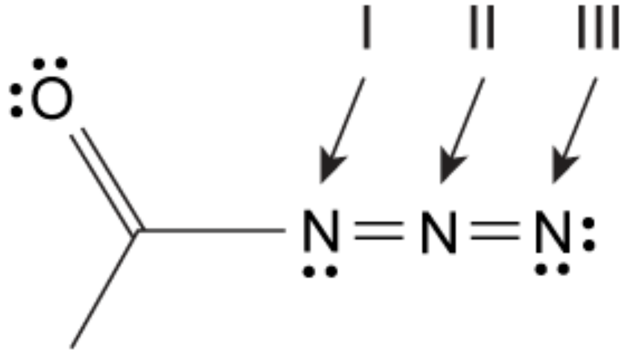
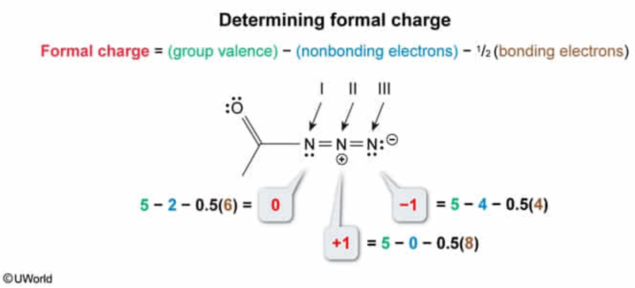
What are the correct formal charges for nitrogen atoms I, II, and III, respectively, in the azide below?



- A. 0, +1, −1 [87%]
- B. +1, 0, −1 [5%]
- C. 0, −1, +1 [4%]
- D. −1, +1, 0 [2%]

Correct
88% answered correctly

Explanation:



A **formal charge** is the charge assigned to an atom based on an accounting method which assumes that the bonding electrons between two atoms are shared equally. Accordingly, in the Lewis structure of a molecule, any nonbonding (lone pair) electrons of an atom are considered to be retained by the atom, whereas bonding electrons are evenly divided between two bonded atoms. The allocated electrons are then assessed relative to an atom's group valence, which is the number of valence electrons held by an uncharged atom within a given group. This is expressed mathematically as:

Formal charge = (group valence) − (nonbonding electrons) − 0.5(bonding electrons)

Examining the given azide structure, the **group valence** for a nitrogen atom is 5 because neutral atoms in Group 15 (5A) have five valence electrons. Formal charge differences between the three nitrogen atoms then relate to the number of bonds and lone pair electrons in each. For example, nitrogen atom II has no **nonbonding electrons** but has 8 **bonding electrons**. Because these electrons are considered to be evenly shared with the other two nitrogen atoms, nitrogen atom II is allocated 4 of the bonding electrons and is deemed to be electron-deficient relative to a neutral valence of 5 electrons. Therefore, nitrogen atom II is assessed a +1 formal charge. Nitrogen atoms I and III are evaluated similarly to give formal charges of zero and −1, respectively.

(Choices B, C, and D) Nitrogen atoms with three bonds and one lone pair have a formal charge of zero (nitrogen atom I). Adding one more bond and removing one lone pair increases the formal charge one unit to +1 (nitrogen atom II). Removing a bond and adding one lone pair decreases the formal charge one unit to −1 (nitrogen atom III).

Educational objective:
A formal charge is the charge assigned to an atom based on an accounting method which assumes that the bonding electrons between two atoms are shared equally. Formal charge equals the group valence of an atom minus the number of nonbonding electrons and minus half of the number of bonding electrons.

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